Manwe Castro

manwe123@asu.edu | (602) 350-3164 | linkedin.com/in/manwe-castro | https://manwe-123.github.io

Education

Arizona State University Tempe, AZ
Bachelor of Science in Mechanical Engineering | GPA: 3.61 *May 2028*

- Dean's List (Fall 2024, Spring 2025, Fall 2025)
- Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Statics, Thermodynamics, Dynamics, Circuits

Marcos de Niza High School Tempe, AZ
Diploma | GPA: 3.54 *May 2024*

- Extracurriculars: Chamber Orchestra (8 years), Varsity Team Captain for Football, Baseball, and Track & Field

Job Experience

Institute for Digital Inclusion Acceleration Chandler, AZ
Digital Navigator *January 2025 – Present*

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.
- Mentor community members on digital literacy and device safety, translating complex technical concepts for users with varying levels of technical proficiency.

Wingstop Tempe, AZ
Cook *June 2025 – August 2025*

- Maintained strict food safety and quality standards in a high-volume, fast-paced kitchen environment.
- Executed precise order fulfillment and managed station prep to ensure rapid service times during peak operational hours.

Sonic Drive-In Tempe, AZ
Shift Lead *April 2022 – August 2024*

- Directed worker efficiency and food preparation in a high-volume, time-sensitive environment, ensuring strict adherence to safety and quality standards.
- Delivered accurate orders and managed customer relations, achieving a 103% customer satisfaction rating.
- Trained new crew members on operational procedures, equipment maintenance, and rapid service protocols.

Tilly's Tempe, AZ
Sales Associate *June 2024 – August 2024*

- Recognized as Employee of the Month (July 2024) for exceeding sales targets and maintaining high store standards.
- Increased sales through effective visual merchandising and personalized customer service.
- Trained new associates on point-of-sale systems and inventory management, enhancing team collaboration and performance.

AZCEND Chandler, AZ
Volunteer *September 2023 – February 2026*

- Managed program data entry and tracking to help leadership measure community impact and secure future funding.
- Coordinated logistics and physical setup for large-scale community resource distributions, including Operation Santa.
- Organized the layout of branch facilities, combining team management with physical labor to optimize service delivery.

Project Experience

AI-Driven Quantitative Equity Research & Portfolio Optimization Terminal *May 2026 – Present*

- Architected a modular, asynchronous Python application integrating multiple external APIs (OpenBB, NewsAPI, Sophtron) to automate the ingestion, normalization, and synchronization of large-scale datasets, creating a low-latency centralized dashboard.
- Implemented custom web scrapers and deployed a local Large Language Model (Ollama/Llama 3.1) to parse SEC filings and synthesize unstructured news sentiment, creating a closed-loop automated pipeline for predictive risk analysis.

- Engineered a 5-factor quantitative scoring model and integrated PyPortfolioOpt to perform mathematical optimization, transforming raw algorithmic signals into actionable, constraint-based recommendations to demonstrate proficiency in complex backend logic.

VR Rapid Prototyping Application Tempe, AZ
Prototyping & Documentation Lead *March 2025 – December 2025*

- Engineered a virtual reality prototyping environment for Meta Quest headsets, scripting interactions in C# and integrating over 100 interactive 3D assets with simulated physics and collision mechanics to support 50 to 75 minute hands-on design sessions across 30+ donated units.
- Directed a cross-functional modeling team by establishing standardized Unity workflows and resolving engine-level rendering blockers, optimizing frame rates to mitigate VR motion sickness.
- Collaborated with faculty to integrate a 2-week curriculum into the experience and developed in-game analytics scripts to track completion time, errors, and hints, creating a data-driven feedback loop for instructors.
- Authored Gantt charts, risk matrices, and standardized technical reports for a 12-person, multi-semester project, delivering a transition package that enabled future teams to continue development, testing, and classroom deployment.

Mental Health Robotics System for Isolated Environments Tempe, AZ
Design and Prototyping Lead *August 2024 – December 2024*

- Programmed an Arduino Uno to process ultrasonic and photoresistive sensor inputs, controlling drive motors and a servo actuator to enable autonomous user tracking and automated medical dispensing, achieving 95%+ proximity detection reliability, repeatable 15 cm stopping distance, and drawer activation below a 25 Lux light value threshold.
- Modeled a 1.8 cubic foot robotic enclosure in SolidWorks and fabricated the structure from laser-cut balsa and plywood, integrating sensors, motors, display hardware, and battery components into a prototype build, finishing 40% under the $75 budget while maintaining lightweight structural rigidity.
- Executed formal acceptance testing across functional requirements, diagnosing and resolving an electrical power delivery fault to pass sensor, motor, and dispensing tests, validating semi-autonomous navigation and automated medical access.

Shock Absorber Assembly *August 2025 – December 2025*

- Designed a fully constrained automotive shock absorber assembly in SolidWorks by modeling custom components, integrating McMaster-Carr standard hardware, and amending initial dimensions to eliminate physical interferences, ensuring proper mechanical fit and material selection for dynamic suspension applications.
- Generated fully dimensioned 2D engineering drawings following ASME Y14.5 standards and 3rd angle projection, detailing complex features such as shaft thread specifications, keyways, and spring parameters to provide machinists with precise, unambiguous fabrication instructions.
- Authored a comprehensive Bill of Materials and exploded assembly drawing to document custom and standard hardware components, establishing a standardized technical reference that streamlines procurement and future mechanical assembly.

Computer-Aided Design Modeling Projects *August 2024 – December 2025*

- Developed SolidWorks and Fusion 360 part files and produced fully dimensioned engineering drawing sheets that follow standard drafting conventions, reinforcing skills in technical communication and manufacturability.
- Applied CAD best practices and design intent principles to strengthen 3D visualization abilities and prepare for advanced coursework in mechanical design and manufacturing.

Skills

CAD & Design: SolidWorks, Autodesk Fusion 360, 2D Drafting, ASME Y14.5, GD&T, MATLAB
Fabrication & Hands-On: FDM 3D Printing, Laser Cutting, Orca Slicer, Circuit Wiring, Automotive Repair, Construction, Roofing
Programming & Controls: Python, Arduino (C++), Unity Game Engine, Git, Sensor Integration, Streamlit, Asynchronous Data Pipelines
Personal Pursuits: Boxing, Salsa Dancing, Sewing and Garment Customization, Rock Climbing, Photography, Music (Violin, Guitar, Piano, Jarana, Requinto)

Study Abroad

EPICS in Spain *Spring 2026*

- Completed a global intensive design program focused on human-centered engineering and smart city infrastructure.
- Partnered with local organizations to conduct door-to-door needs assessments for elderly residents, applying human-centered design principles to community challenges.
- Analyzed urban infrastructure through guided smart city tours and evaluated community impact models at local social gardens.
- Immersed in local culture through flamenco instruction and historical site analysis to understand the intersection of cultural heritage and modern urban systems.

Activities and Campus Involvement

Society of Hispanic Professional Engineers Tempe, AZ
Outreach Director *March 2026 - Present*

- Direct chapter outreach by partnering with local schools and community organizations, prioritizing schools with large Hispanic student populations.
- Plan and coordinate 8+ outreach and professional development events per year, including Noche de Ciencias and Engineers Week.
- Manage a 4-member student events committee, 15+ volunteers, and a $500 annual budget while designing hands-on technical workshops that build practical engineering skills.
- Served as Marketing Chair on the Finance Committee (January 2026 to March 2026), supporting event promotion and internal chapter operations.

Boxing at ASU Tempe, AZ
General Member *January 2025 - Present*

Salsa at ASU Tempe, AZ
General Member *January 2026 - Present*

Climbing at ASU Tempe, AZ
General Member *August 2026 - Present*

Certifications/Awards

Dean's List Fall 2024, Spring 2025, Fall 2025

- Maintained above a 3.5 GPA while taking 12+ credit hours